UNIT 1 INTRODUCTION **9 Hrs.**

Introduction to Big Data - Challenges of Conventional Systems - Nature of Data - Small data - Medium data - Big Data - Small data vs Big data - Sources of Big Data - Big Data Characteristics - Big Data Analytics - Importance of Big Data, Big Data in the Enterprise - Big Data Enterprise Model - Building a Big Data Platform - Big data in Social and Behavioral sciences.

Introduction

What is Data?

Data can be defined as a representation of facts, concepts, or instructions in a formalized manner.

Table 1.1 Characteristics of Data

Accuracy	Is the information correct in every detail?
Completeness	How comprehensive is the information?
Reliability	Does the information contradict other trusted resources?
Relevance	Do you really need this information?
Timeliness	How up- to-date is information? Can it be used for real-time reporting?

Differences between Small Data, Medium Data and Big Data

Data can be small, medium or big.

Small data is data in a volume and format that makes it accessible, informative and actionable.

Medium data refers to data sets that are too large to fit on a single machine but don't require enormous clusters of thousands.

Big data is extremely large data sets that may be analysed computationally to reveal patterns, trends, and associations, especially relating to human behaviour and interactions.

Table 1.2 Small Data and Big Data Comparison Table

Basis of Comparison	Small Data	Big Data
Definition	Data that is 'small' enough for human .In a volume and format that makes it accessible, informative and actionable	Data sets that are so large or complex that traditional data processing applications cannot deal with them
Data Source	● Data from traditional enterprise systems like	● Purchase data from point-of-sale

	○ Enterprise resource planning ○ Customer relationship management(CRM)	● Clickstream data from websites ● GPS stream data – Mobility data sent to a server ● Social media – Facebook, Twitter
Volume	Most cases in a range of tens or hundreds of GB.Some case few TBs (1 TB=1000 GB)	More than a few Terabytes (TB)
Velocity (Rate at which data appears)	● Controlled and steady data flow ● Data accumulation is slow	● Data can arrive at very fast speeds. ● Enormous data can accumulate within very short periods of time
Variety	Structured data in tabular format with fixed schema and semi-structured data in JSON or XML format	High variety data sets which include Tabular data,Text files, Images, Video, Audio, XML,JSON,Logs,Sensor data etc.
Veracity (Quality of data)	Contains less noise as data collected in a controlled manner.	Usually, the quality of data not guaranteed. Rigorous data validation is required before processing.
Value	Business Intelligence, Analysis, and Reporting	Complex data mining for prediction, recommendation, pattern finding, etc.
Time Variance	Historical data equally valid as data represent solid business interactions	In some cases, data gets older soon(Eg fraud detection).
Data Location	Databases within an enterprise, Local servers, etc.	Mostly in distributed storages on Cloud or in external file systems.
Infrastructure	Predictable resource allocation.Mostly vertically scalable hardware	More agile infrastructure with a horizontally scalable architecture. Load on the system varies a lot.

Introduction to Big Data

Big data is data that exceeds the processing capacity of conventional database systems. The data is too big, moves too fast, or doesn't fit the strictures of your database architectures. To gain value from this data, you must choose an alternative way to process it. Big Data has to deal with large and complex datasets that can be structured, Semi-structured, or unstructured and will typically not fit into memory to be Processed.

Big data is a field that treats ways to analyze, systematically extract information from, or otherwise deal with data sets that are too large or complex to be dealt with by traditional data-processing application software. – Wikipedia

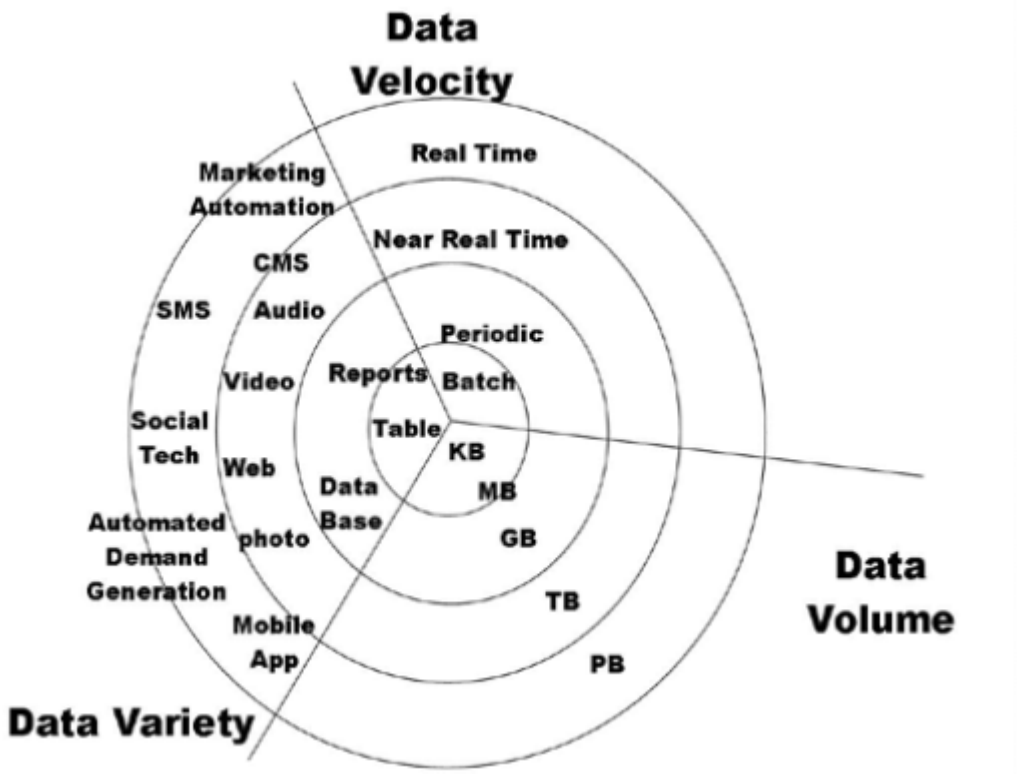

Fig. 1.1 3Vs of Big Data

Examples of Big Data:

The New York Stock Exchange generates about one terabyte of new trade data per day.

The statistic shows that 500+terabytes of new data get ingested into the databases of social media site Facebook, every day. This data is mainly generated in terms of photo and video uploads, message exchanges, putting comments etc.

A single Jet engine can generate 10+terabytes of data in 30 minutes of flight time. With many thousand flights per day, generation of data reaches up to many Petabytes.

Table 1.3 Examples of Data Volumes

Unit	Value	Example
Kilobytes (KB)	1,000 bytes	a paragraph of a text document
Megabytes (MB)	1,000 Kilobytes	a small novel
Gigabytes (GB)	1,000 Megabytes	Beethoven's 5th Symphony
Terabytes (TB)	1,000 Gigabytes	all the X-rays in a large hospital
Petabytes (PB)	1,000 Terabytes	half the contents of all US academic research libraries
Exabytes (EB)	1,000 Petabytes	about one fifth of the words people have ever spoken
Zettabytes (ZB)	1,000 Exabytes	as much information as there are grains of sand on all the world's beaches

Yottabytes (YB)	1,000 Zettabytes	as much information as there are atoms in 7,000 human bodies

Advantages of using Big Data

1. Improved business processes
2. Fraud detection
3. Improved customer service
4. Better decision-making
5. Increased productivity
6. Reduce costs
7. Improved customer service
8. Fraud detection
9. Increased revenue
10. Increased agility
11. Greater innovation
12. Faster speed to market

Disadvantages of Big Data

1. Privacy and security concerns
2. Need for technical expertise
3. Need for talent
4. Data quality
5. Need for cultural change
6. Compliance
7. Cybersecurity risks
8. Rapid change
9. Hardware needs
10. Costs
11. Difficulty integrating legacy systems